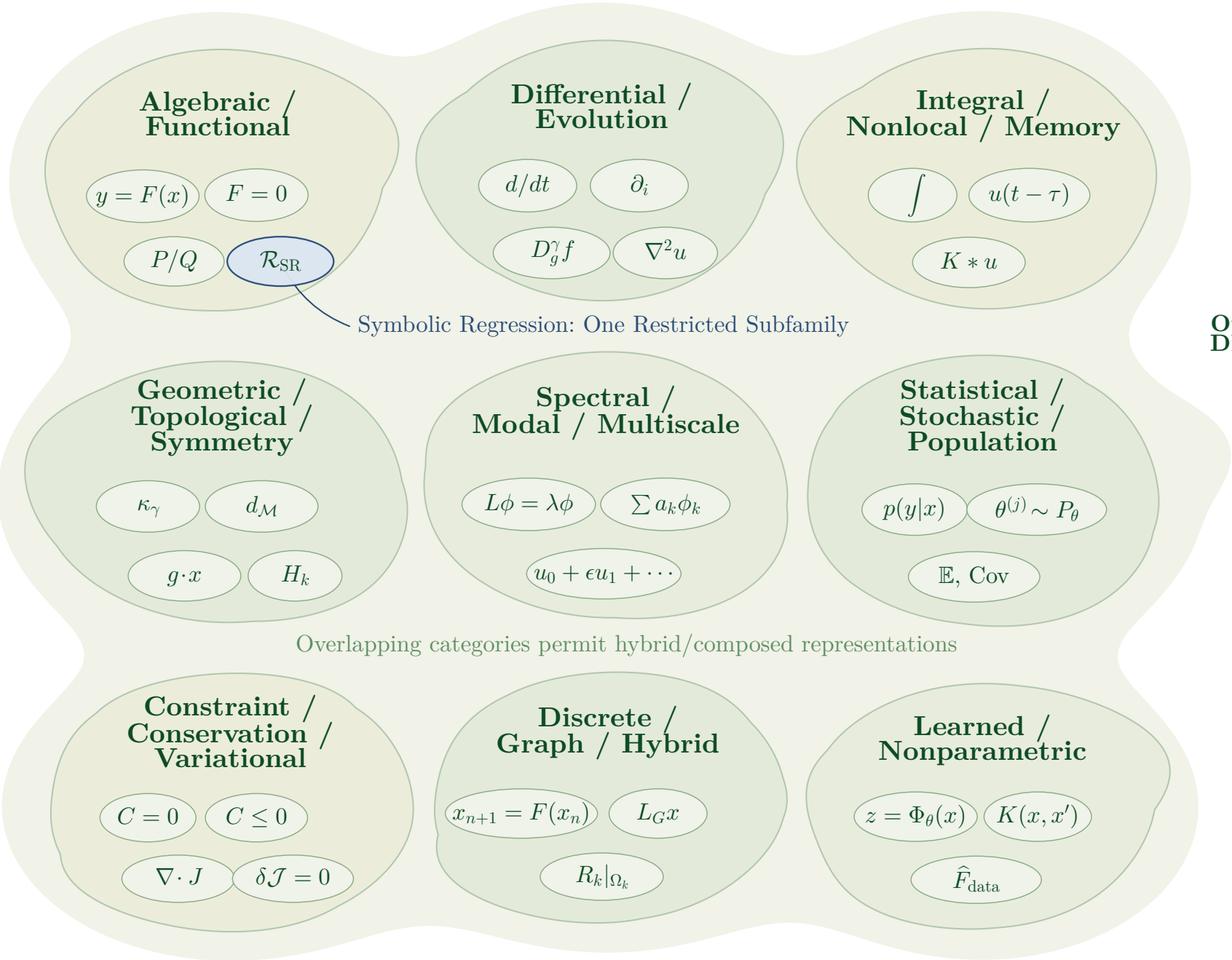


Admissible Relational Space $\mathfrak{R}_q$



Illustrative, non-exhaustive categories; each SIR run instantiates a finite, auditable grammar.

Coordinate-Generation Operators

Representative groups; recursively composable

- $\partial, d, \Delta, D^\gamma$

Local
Derivatives, differences,
tangent / phase constructions
- $\int, *, \mathcal{L}_\tau$

Nonlocal / History
Integrals, convolutions,
lags and memory operators
- $\mathcal{T}, \mathcal{B}, \Pi$

Algebraic / Basis
Transforms, ratios, reductions,
nondimensional groups and bases
- $\mathcal{G}, \mathcal{S}$

Structural / Statistical
Geometry, symmetry, graph,
spectral and statistical maps
- Φ_θ

Learned
Embeddings, kernels,
neural / surrogate coordinates
- $\mathcal{M}$

Model Primitives
Numerical simulators, pretrained
models, scientific codes, agents

Open, Expandable
Discovery Frontier



$$G = G_{i_k} \circ \cdots \circ G_{i_2} \circ G_{i_1}$$

Operators may compose recursively

Key  Category  Illustrative subfamily